The book walkthrough: one command, then Excel

What the firm asked for, 14 Oct 2026: *"i want you to make it work so i can generate this workbook at work with the script, it should be debugged and walked through and tied out."*

This is the walk: the workbook built the way it is built at work, then used in Excel 16 the way a person uses it (clicking the buttons, typing in the yellow row, saving, closing and opening it again), and every answer read back from Excel and set beside an answer written down beforehand from the source files.

1. Build

From `threshold-engine/`:

```
python -m threshold_engine peers pull --cache books/ind_cache --first 2006Q1 --last 2026Q2 --products "Cre
dit card" Mortgage "Home equity" "Other consumer" Auto --delinquency yes --mergers 2005-01-01
python -m threshold_engine book books/consumer-public.json consumer-credit.xlsx
```

The walkthrough ran the second command on a copy of the book whose only change is where the cache is (`prepare.py`): the cache pulled on 14 Oct 2026 was used rather than pulled again. Expected: `Wrote consumer-credit.xlsx: 105 series, 82 quarters; open it on Summary, decide on Decide.` It took 7 min 49 s on the last run (12 min 42 s on the first), almost all of it the peers build reading the cache.

2. The steps

The steps are cumulative, as a session is: what is typed in one step is still there in the next.

Step	What you do	What it shows
B1	Open the workbook	It opens on Summary; the tabs in order (Assess, Chart and Peers workings hidden); three slicers, one button each; Decide shows the first series; nothing decided; no links, no errors
B2	Read Summary	All 105 series' latest value and rating equal the reference
B3	On Decide, click Credit card, Nonaccrual, KeyBank	One row; the headline, the lines in words, the Check, and the same measure from the industry and peers
B4	Type 3.5 under How rare High is (z)	Only High's line moves; Thresholds and Summary follow
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	The lines move as the reference says; Assess lists the period, whole
B6, B6b	Clear the reason; type it back	Refused, in words, on Decide; then the lines as in B5
B7	Accept these lines, with a note	Decided: 1 of 105
B8	Click Auto, Net charge-offs, KeyBank	Life to date: 15 quarters, its lines and its Check
B9	Click Mortgage, Net charge-offs, Industry	Refused, with the number that would fix it
B10	Save, close, open again	Every decision and setting still there, and the lines with them

A step passes when every cell it reads says what `steps.json` says: a number within 1e-9, text exactly.

3. How it was run

```
python docs/walkthrough-book/prepare.py PATH-TO-CACHE
python -m threshold_engine book docs/walkthrough-book/out/book.json docs/walkthrough-book/out/consumer-cre
dit.xlsx
python docs/walkthrough-book/expected.py docs/walkthrough-book/out/book.json
copy docs\walkthrough-book\out\consumer-credit.xlsx docs\walkthrough-book\out\walk.xlsx
powershell -NoProfile -ExecutionPolicy Bypass -File docs\walkthrough-book\walk.ps1
python docs/walkthrough-book/compare.py
```

`expected.py` writes the answers from the series CSVs and `scale.py`, never from the workbook. `walk.ps1` does each step in Excel through Excel's own automation and reads each answer back. `compare.py` writes `RESULT.md`.

And, separately, that the file opens without Excel repairing it:

```
powershell -NoProfile -ExecutionPolicy Bypass -File tools\check_slicers_in_excel.ps1 -Path docs\walkthroug
h-book\out\consumer-credit.xlsx -Sheet Decide -Table Decisions -Shapes 4 -Expect "Slicer_Product=Credit ca
rd|Slicer_Measure=Net charge-offs|Slicer_Source=Industry (every bank)" -SwitchCache Slicer_Source -SwitchT
o KeyBank
```

"No errors" in B1 means none a reader can see: on the visible tabs, outside hidden rows and columns. The rest are counted by kind and are all #N/A, the gap a chart leaves where a quarter has no value (on the run of 15 Oct 2026: Assess 169, Chart 8,470, Decide 249, Peers workings 1,556).

4. What it does not prove

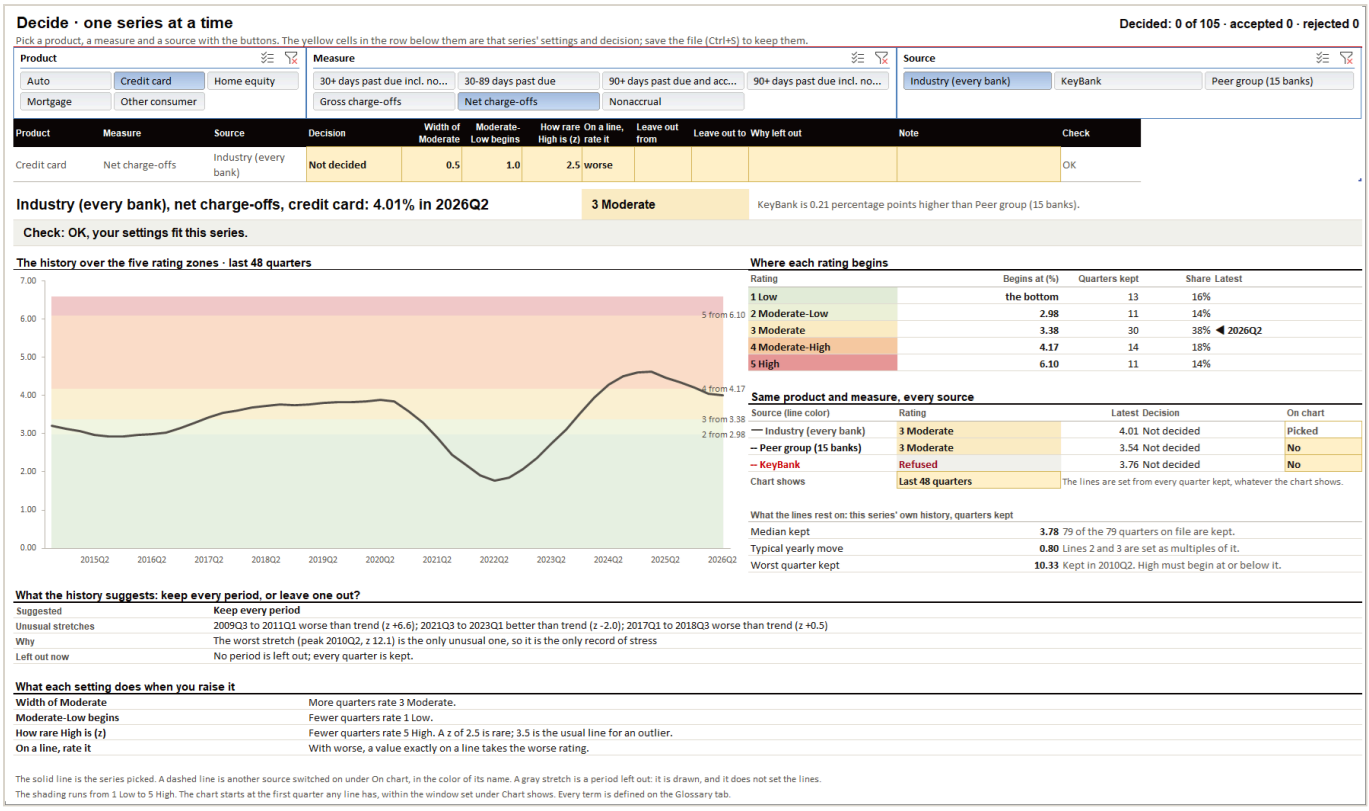
- The peer tabs are counted for errors and links (B1) but their numbers are not checked against a reference here; `tests/test_peers*.py` hold them.
- One series is tied to the filing (`../tie-out/keybank-card-nonaccrual-2026-10-14/`); the other 104 rest on the same code and the reference, not on a filing each.
- Thresholds' latest value is written by the build, not a live formula on Data: pasting a corrected quarter into Data moves the lines but not the latest value (found by the tie-out).

5. The result

327 of 328 reads as expected.

Step	What you do	Reads as expected
B1	Open the workbook as built	8 of 9
B2	Read Summary	210 of 210
B3	On Decide, click Credit card, Nonaccrual, KeyBank	26 of 26
B4	Type 3.5 under How rare High is (z), in the yellow row	13 of 13
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	15 of 15
B6	Clear the reason	1 of 1
B6b	Type the reason back	12 of 12
B7	Pick Accept these lines and write a note	2 of 2
B8	Click Auto, Net charge-offs, KeyBank	12 of 12
B9	Click Mortgage, Net charge-offs, Industry	10 of 10
B10	Save (Ctrl+S), close Excel, open the file again	18 of 18

6. What Excel showed



Decide · one series at a time — Decided: 0 of 105 · accepted 0 · rejected 0

Pick a product, a measure and a source with the buttons. The yellow cells in the row below them are that series' settings and decision; save the file (Ctrl+S) to keep them.

Product: Auto | Credit card | Home equity | Mortgage | Other consumer

Measure: 30+ days past due incl. no... | 30-89 days past due | 90+ days past due and acc... | 90+ days past due incl. no... | Gross charge-offs | Net charge-offs | Nonaccrual

Source: Industry (every bank) | KeyBank | Peer group (15 banks)

Product	Measure	Source	Decision	Width of Moderate	Moderate-Low begins	How rare High is (z)	On a line, rate it	Leave out from	Leave out to	Why left out	Note	Check
Credit card	Net charge-offs	Industry (every bank)	Not decided	0.5	1.0	2.5	worse					OK

Industry (every bank), net charge-offs, credit card: 4.01% in 2026Q2 — 3 Moderate — KeyBank is 0.21 percentage points higher than Peer group (15 banks).

Check: OK, your settings fit this series.

The history over the five rating zones · last 48 quarters

Where each rating begins

Rating	Begins at (%)	Quarters kept	Share	Latest
1 Low	the bottom	13	16%	
2 Moderate-Low	2.98	11	14%	
3 Moderate	3.38	30	38%	◀ 2026Q2
4 Moderate-High	4.17	14	18%	
5 High	6.10	11	14%	

Same product and measure, every source

Source (line color)	Rating	Latest	Decision	On chart
— Industry (every bank)	3 Moderate	4.01	Not decided	Picked
-- Peer group (15 banks)	3 Moderate	3.54	Not decided	No
– KeyBank	Refused	3.76	Not decided	No
Chart shows	Last 48 quarters	The lines are set from every quarter kept, whatever the chart shows.		

What the lines rest on: this series' own history, quarters kept

Median kept	3.78	79 of the 79 quarters on file are kept.
Typical yearly move	0.80	Lines 2 and 3 are set as multiples of it.
Worst quarter kept	10.33	Kept in 2010Q2. High must begin at or below it.

What the history suggests: keep every period, or leave one out?

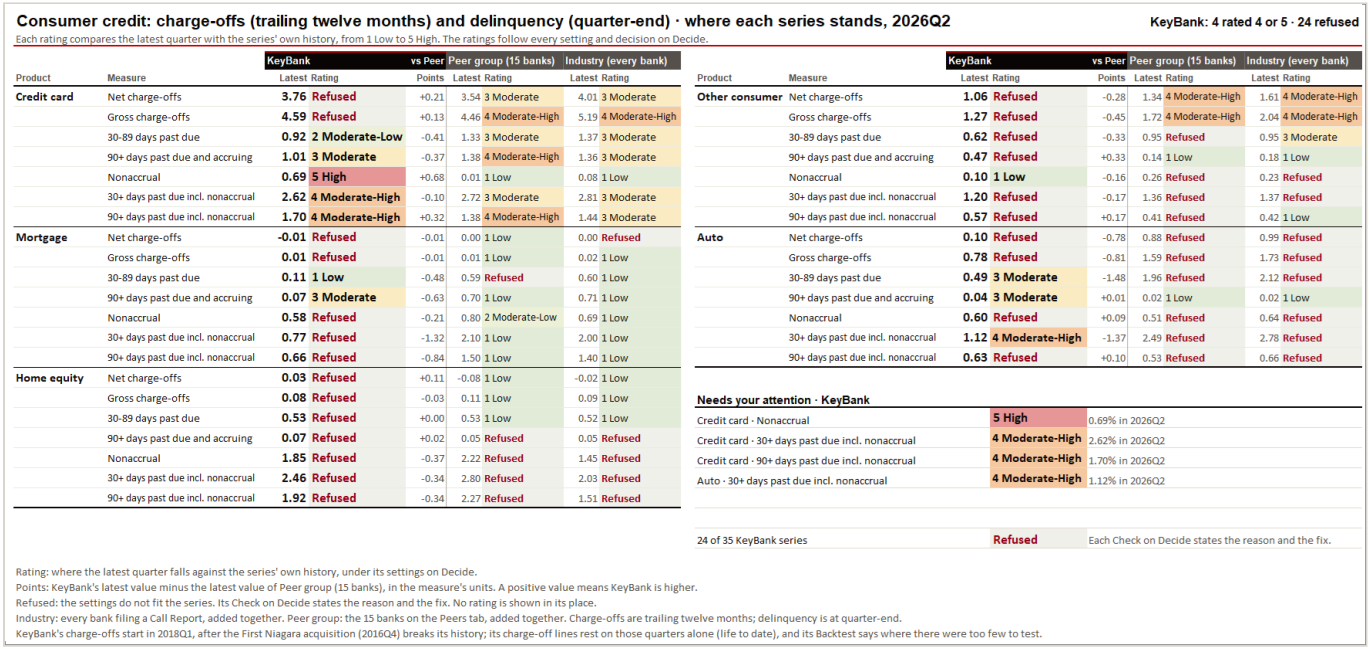
Suggested	Keep every period
Unusual stretches	2009Q3 to 2011Q1 worse than trend (z +6.6); 2021Q3 to 2023Q1 better than trend (z -2.0); 2017Q1 to 2018Q3 worse than trend (z +0.5)
Why	The worst stretch (peak 2010Q2, z 12.1) is the only unusual one, so it is the only record of stress
Left out now	No period is left out; every quarter is kept.

What each setting does when you raise it

Width of Moderate	More quarters rate 3 Moderate.
Moderate-Low begins	Fewer quarters rate 1 Low.
How rare High is (z)	Fewer quarters rate 5 High. A z of 2.5 is rare; 3.5 is the usual line for an outlier.
On a line, rate it	With worse, a value exactly on a line takes the worse rating.

The solid line is the series picked. A dashed line is another source switched on under On chart, in the color of its name. A gray stretch is a period left out: it is drawn, and it does not set the lines.
The shading runs from 1 Low to 5 High. The chart starts at the first quarter any line has, within the window set under Chart shows. Every term is defined on the Glossary tab.

B1 · Open the workbook as built (Decide tab, photographed by Excel during the walk)



Consumer credit: charge-offs (trailing twelve months) and delinquency (quarter-end) · where each series stands, 2026Q2 — KeyBank: 4 rated 4 or 5 · 24 refused

Each rating compares the latest quarter with the series' own history, from 1 Low to 5 High. The ratings follow every setting and decision on Decide.

Product	Measure	KeyBank Latest	KeyBank Rating	vs Peer Points	Peer group (15 banks) Latest	Peer group (15 banks) Rating	Industry (every bank) Latest	Industry (every bank) Rating
Credit card	Net charge-offs	3.76	Refused	+0.21	3.54	3 Moderate	4.01	3 Moderate
	Gross charge-offs	4.59	Refused	+0.13	4.46	4 Moderate-High	5.19	4 Moderate-High
	30-89 days past due	0.92	2 Moderate-Low	-0.41	1.33	3 Moderate	1.37	3 Moderate
	90+ days past due and accruing	1.01	3 Moderate	-0.37	1.38	4 Moderate-High	1.36	3 Moderate
	Nonaccrual	0.69	5 High	+0.68	0.01	1 Low	0.08	1 Low
	30+ days past due incl. nonaccrual	2.62	4 Moderate-High	-0.10	2.72	3 Moderate	2.81	3 Moderate
	90+ days past due incl. nonaccrual	1.70	4 Moderate-High	+0.32	1.38	4 Moderate-High	1.44	3 Moderate
Mortgage	Net charge-offs	-0.01	Refused	-0.01	0.00	1 Low	0.00	Refused
	Gross charge-offs	0.01	Refused	-0.01	0.01	1 Low	0.02	1 Low
	30-89 days past due	0.11	1 Low	-0.48	0.59	Refused	0.60	1 Low
	90+ days past due and accruing	0.07	3 Moderate	-0.63	0.70	1 Low	0.71	1 Low
	Nonaccrual	0.58	Refused	-0.21	0.80	2 Moderate-Low	0.69	1 Low
	30+ days past due incl. nonaccrual	0.77	Refused	-1.32	2.10	1 Low	2.00	1 Low
	90+ days past due incl. nonaccrual	0.66	Refused	-0.84	1.50	1 Low	1.40	1 Low
Home equity	Net charge-offs	0.03	Refused	+0.11	-0.08	1 Low	-0.02	1 Low
	Gross charge-offs	0.08	Refused	-0.03	0.11	1 Low	0.09	1 Low
	30-89 days past due	0.53	Refused	+0.00	0.53	1 Low	0.52	1 Low
	90+ days past due and accruing	0.07	Refused	+0.02	0.05	Refused	0.05	Refused
	Nonaccrual	1.85	Refused	-0.37	2.22	Refused	1.45	Refused
	30+ days past due incl. nonaccrual	2.46	Refused	-0.34	2.80	Refused	2.03	Refused
	90+ days past due incl. nonaccrual	1.92	Refused	-0.34	2.27	Refused	1.51	Refused

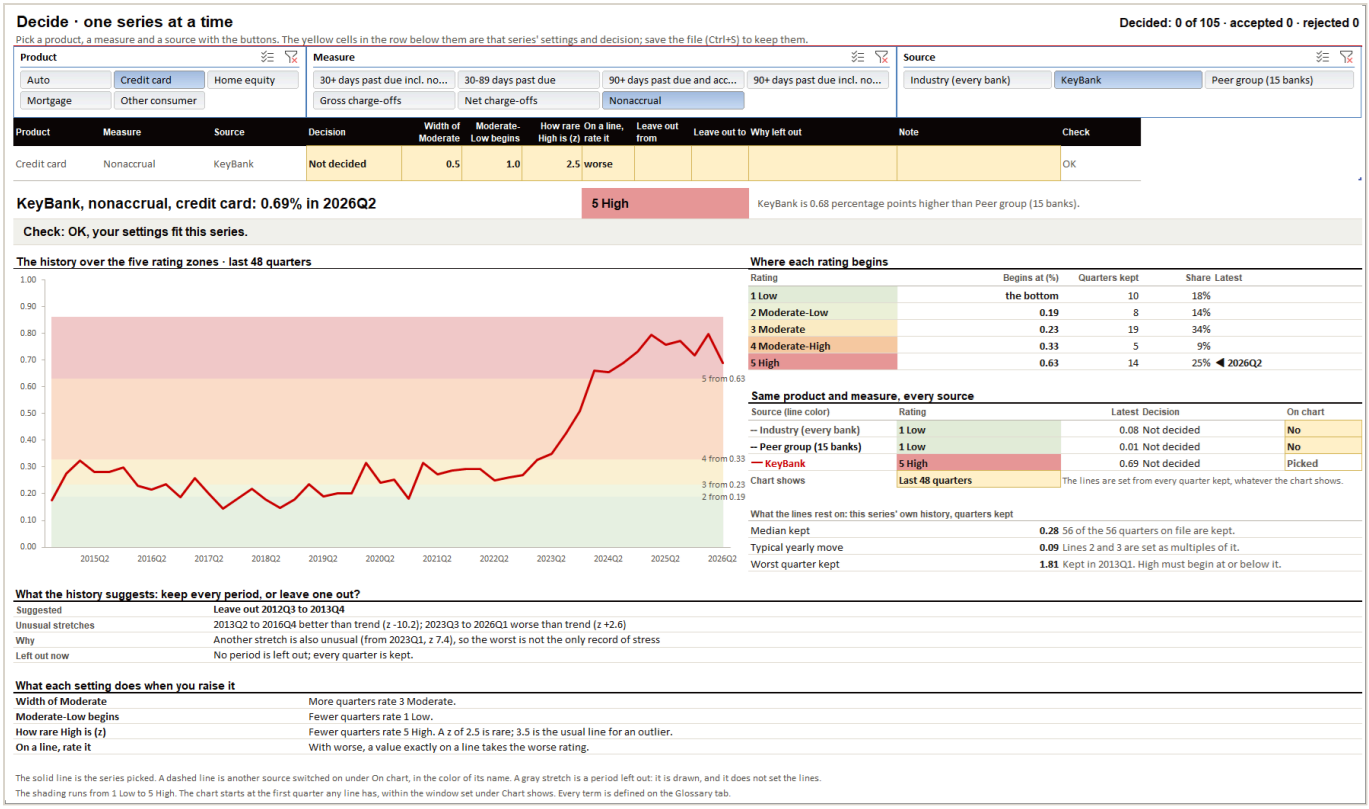
Product	Measure	KeyBank Latest	KeyBank Rating	vs Peer Points	Peer group (15 banks) Latest	Peer group (15 banks) Rating	Industry (every bank) Latest	Industry (every bank) Rating
Other consumer	Net charge-offs	1.06	Refused	-0.28	1.34	4 Moderate-High	1.61	4 Moderate-High
	Gross charge-offs	1.27	Refused	-0.45	1.72	4 Moderate-High	2.04	4 Moderate-High
	30-89 days past due	0.62	Refused	-0.33	0.95	Refused	0.95	3 Moderate
	90+ days past due and accruing	0.47	Refused	+0.33	0.14	1 Low	0.18	1 Low
	Nonaccrual	0.10	1 Low	-0.16	0.26	Refused	0.23	Refused
	30+ days past due incl. nonaccrual	1.20	Refused	-0.17	1.36	Refused	1.37	Refused
	90+ days past due incl. nonaccrual	0.57	Refused	+0.17	0.41	Refused	0.42	1 Low
Auto	Net charge-offs	0.10	Refused	-0.78	0.88	Refused	0.99	Refused
	Gross charge-offs	0.78	Refused	-0.81	1.59	Refused	1.73	Refused
	30-89 days past due	0.49	3 Moderate	-1.48	1.96	Refused	2.12	Refused
	90+ days past due and accruing	0.04	3 Moderate	+0.01	0.02	1 Low	0.02	1 Low
	Nonaccrual	0.60	Refused	+0.09	0.51	Refused	0.64	Refused
	30+ days past due incl. nonaccrual	1.12	4 Moderate-High	-1.37	2.49	Refused	2.78	Refused
	90+ days past due incl. nonaccrual	0.63	Refused	+0.10	0.53	Refused	0.66	Refused

Needs your attention · KeyBank

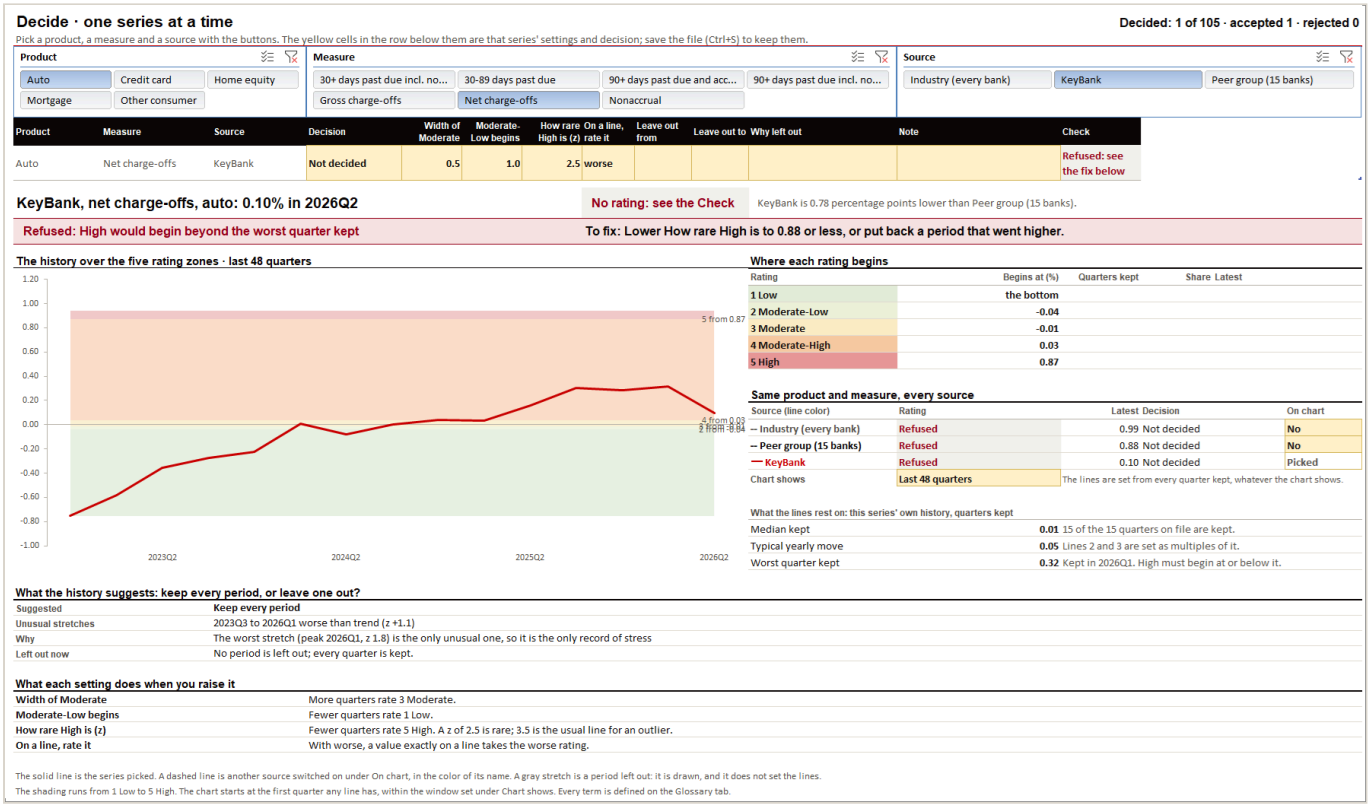
Credit card · Nonaccrual	5 High	0.69% in 2026Q2
Credit card · 30+ days past due incl. nonaccrual	4 Moderate-High	2.62% in 2026Q2
Credit card · 90+ days past due incl. nonaccrual	4 Moderate-High	1.70% in 2026Q2
Auto · 30+ days past due incl. nonaccrual	4 Moderate-High	1.12% in 2026Q2
24 of 35 KeyBank series	Refused	Each Check on Decide states the reason and the fix.

Rating: where the latest quarter falls against the series' own history, under its settings on Decide.
Points: KeyBank's latest value minus the latest value of Peer group (15 banks), in the measure's units. A positive value means KeyBank is higher.
Refused: the settings do not fit the series. Its Check on Decide states the reason and the fix. No rating is shown in its place.
Industry: every bank filing a Call Report, added together. Peer group: the 15 banks on the Peers tab, added together. Charge-offs are trailing twelve months; delinquency is at quarter-end.
KeyBank's charge-offs start in 2018Q1, after the First Niagara acquisition (2016Q4) breaks its history; its charge-off lines rest on those quarters alone (life to date), and its Backtest says where there were too few to test.

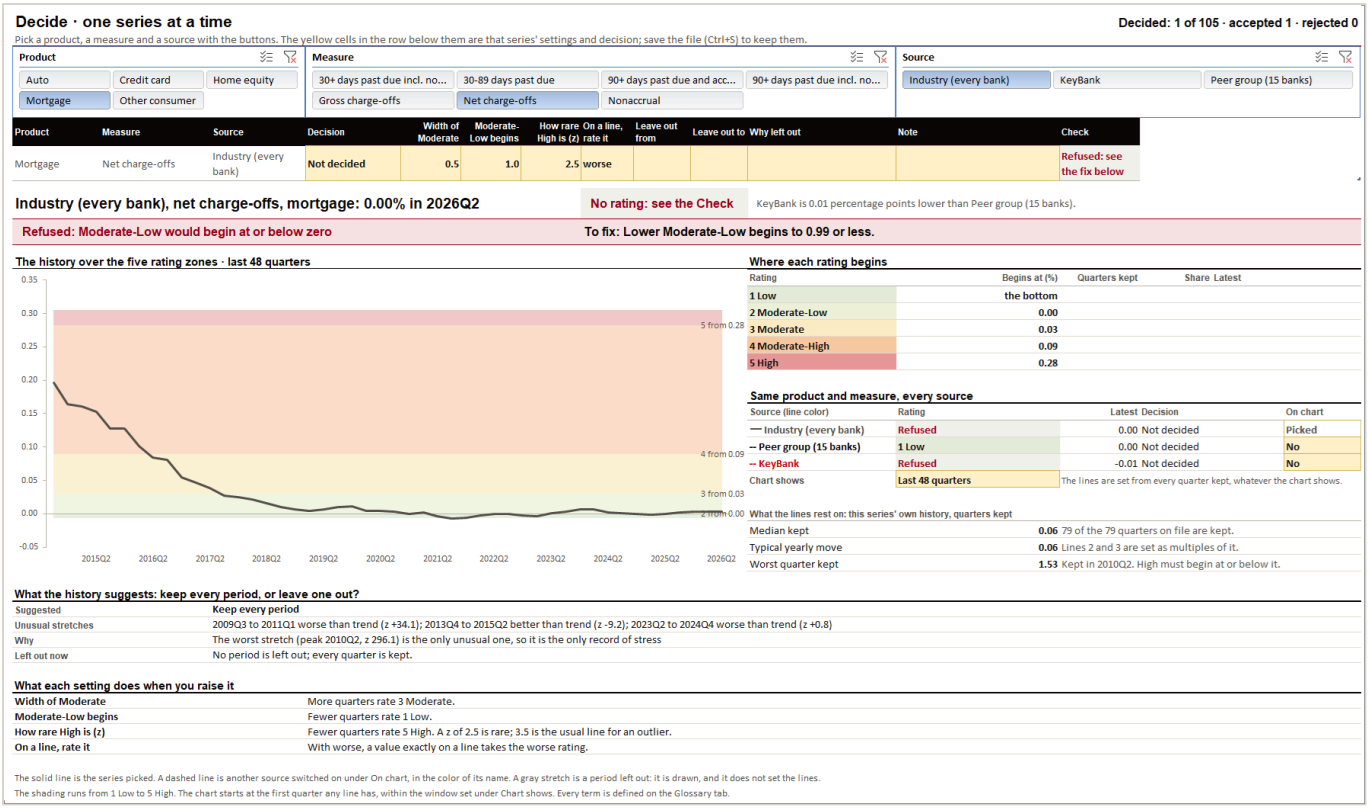
B1 · Open the workbook as built (Summary tab, photographed by Excel during the walk)



B3 · On Decide, click Credit card, Nonaccrual, KeyBank (Decide tab, photographed by Excel during the walk)



B8 · Click Auto, Net charge-offs, KeyBank (Decide tab, photographed by Excel during the walk)



B9 · Click Mortgage, Net charge-offs, Industry (Decide tab, photographed by Excel during the walk)